

Experiments - 1

:- Introduction to Networking Simulation Tools : Cisco Packet Tracer.

:- Cisco Packet Tracer

:- Cisco Packet Tracer can be used to create complicated network topologies, as well as to test and simulate abstract networking concepts. It acts as a playground for you to explore networking and the experience is very close to what you see in computer networks.

Packet Tracer allows you to drag and drop routers, switches and other network devices to create simulated network topologies.

Why use Cisco Packet Tracer?

It provides many features like :-

- Users can configure network devices and simulate network traffic to test and troubleshoot network configurations, which saves time and resources and reduce the time of network downtime.
- It enables its users to simulate the configuration relating to Cisco routers and can be accessed anywhere anytime.

- It provides an interactive and self-paced environment.
- Supports a multi-user system that allows many users to connect various topologies across a computer network. Instructors can also build exercises for students to perform using Packet Tracer.

Working of Cisco Packet Tracer

Here are the basic steps of how Cisco Packet Tracer works:-

1. Launch Packet Tracer on your computer.
2. Create a new network topology by dragging and dropping network devices such as routers, switches and PCs into the workspace.
3. Connect the devices by dragging cables between them.
4. Configure the devices by using the command line interface or the graphical user interface.
5. Test the network by sending packets between devices and verifying connectivity.
6. Troubleshoot any issues that arise by using the built-in tools and features of Packet Tracer.

Types of Wires

4. Coaxial cable:- This type of cable is used for cable television and broadband internet connections. It has a copper core surrounded by insulation, a braided shield and an outer jacket.

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2. Twisted Pair Cable :- This type of cable is used for Ethernet connections and is made up of pairs of wires twisted together to reduce interference.
3. Fibre Optical Cable :- This type of data is used for high speed data transmission over long distances. It uses light to transmit data and is immune to electromagnetic interference.

Fast Ethernet :- Fast Ethernet is a type of Ethernet network that provides data transfer rates up to 100 Mbps. In Cisco Packet Tracer, Fast Ethernet wires can be added to routers by using the appropriate network modules. Fast Ethernet wires can be used to connect devices such as routers, switches and PCs to a network.

Experiment 2

Aim: Design and implement topology of two nodes (Node1, Node2) separated by a point-to-point link specifying bandwidth & delay.

Software Used: Cisco Packet Tracer

Theory:

Network Topology Design:

- A network topology defines the physical or logical arrangement of devices and links in a network.

Point-to-Point Link:

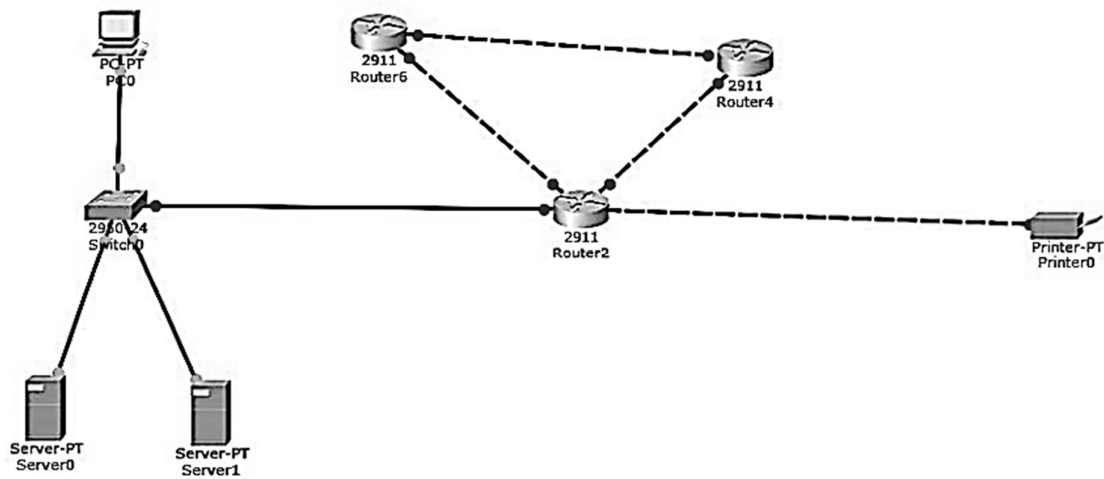
- A point-to-point link is a direct connection between two devices without any intermediate devices, such as switches or routers.
- It is often used for connecting two remote sites or devices.

Bandwidth:

- Bandwidth refers to the maximum data transfer rate of a network link, typically measured in bits per second (bps).
- It determines how much data can be transmitted over the link in a given time.

Delay:

- Delay, often referred to as latency, is the time it takes for a data packet to travel from the source to the destination over a network link.
- It is usually measured in milliseconds (ms).



The network topology shown in the image consists of three routers, two servers, a printer, and a switch. The routers are responsible for routing traffic between different parts of the network. The switch is responsible for connecting the devices to the routers and to each other.

The PC-PT and Printer-PT devices are connected directly to the switch. The Server-PT devices are connected to the routers using dedicated Ethernet interfaces.

This topology is a good example of a mesh network topology. Mesh networks are highly redundant and scalable.

Experiment 3

Aim: To Design and Implement LAN an interconnecting device for two different LANs and to evaluate network performance parameters. Design a topology of P2P (2 node) and CSMA channel (4 nodes) specifying bandwidth 5Mbps and 100Mbps respectively. Study IP addresses assignment mechanism.

Software Used: Cisco Packet Tracer

Theory

To design and implement a LAN interconnecting device for two different LANs, we can use a router. A router is a network device that is used to connect two or more networks together. It can also be used to route traffic between different networks.

LAN

A local area network (LAN) is a computer network that interconnects devices within a limited geographical area, such as a home, office, or school. LANs are typically used to share resources, such as files, printers, and internet access.

LANs can be wired or wireless. Wired LANs use Ethernet cables to connect devices, while wireless LANs use radio waves. Wireless LANs are becoming more popular because they are more convenient and easier to set up.

P2P (2 node)

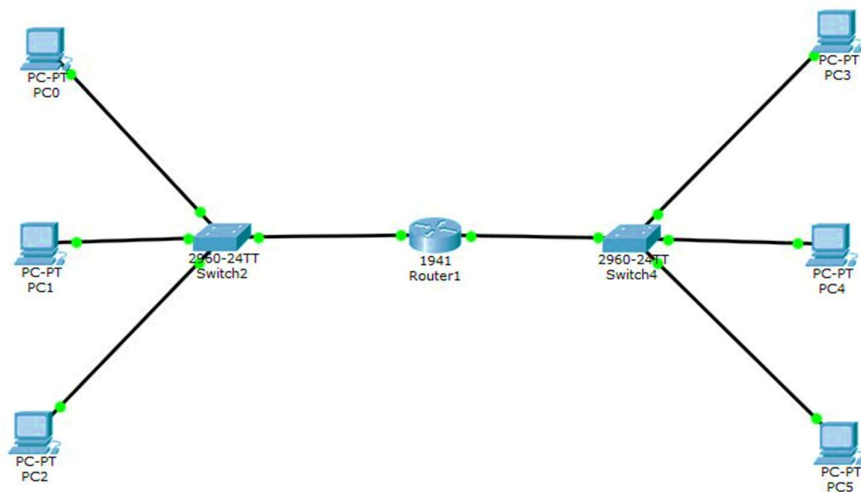
To implement a P2P (2 node) topology, we can simply connect two computers together using a crossover Ethernet cable. A crossover Ethernet cable is a type of Ethernet cable that allows two devices to be connected directly to each other without the need for a switch or router.

CSMA channel (4 nodes)

To implement a CSMA channel (4 nodes) topology, we can connect the four computers together using a hub or a switch. A hub is a network device that connects multiple devices together and allows them to communicate with each other. A switch is a more intelligent network device than a hub and can learn the MAC addresses of the devices that are connected to it. This allows the switch to forward packets directly to the destination device, which can improve network performance.

Bandwidth

The bandwidth of a network is the amount of data that can be transferred over the network in a given period of time. The bandwidth of a P2P (2 node) topology is 5Mbps, and the bandwidth of a CSMA channel (4 nodes) topology is 100Mbps.



Experiment 4

Aim - Conducting a Network Capture and Monitoring with Cisco Packet Tracer

Software Used- Cisco packet tracer

Theory

The steps to design and implement a LAN interconnecting device for two different LANs and evaluate network performance parameters:

1. Connect the PCs to the switches.

- Connect each of the PCs on the left to the 2960-24TT switch on the left using an Ethernet cable.
- Connect each of the PCs on the right to the 2960-24TT switch on the right using an Ethernet cable.

2. Connect the switches to the router.

- Connect the 2960-24TT switch on the left to the router using an Ethernet cable.
- Connect the 2960-24TT switch on the right to the router using an Ethernet cable.

3. Configure the switches.

You do not need to configure the switches for this network. The switches will automatically detect the devices that are connected to them and configure themselves accordingly.

4. Configure the router.

- Assign an IP address to each of the router's LAN ports.
- Configure the router to route traffic between the two LANs.

- Configure the router's DHCP server to assign IP addresses to the PCs on the network.

5. Configure the PCs.

- Assign an IP address to each of the PCs.
- Configure the PCs to use the router as their default gateway.

IP address and gateway configuration:

Router:

- LAN port 1: IP address 192.168.1.1
- LAN port 2: IP address 192.168.2.1

PCs on the left LAN:

- PC 1: IP address 192.168.1.2, gateway 192.168.1.1
- PC 2: IP address 192.168.1.3, gateway 192.168.1.1
- PC 3: IP address 192.168.1.4, gateway 192.168.1.1

PCs on the right LAN:

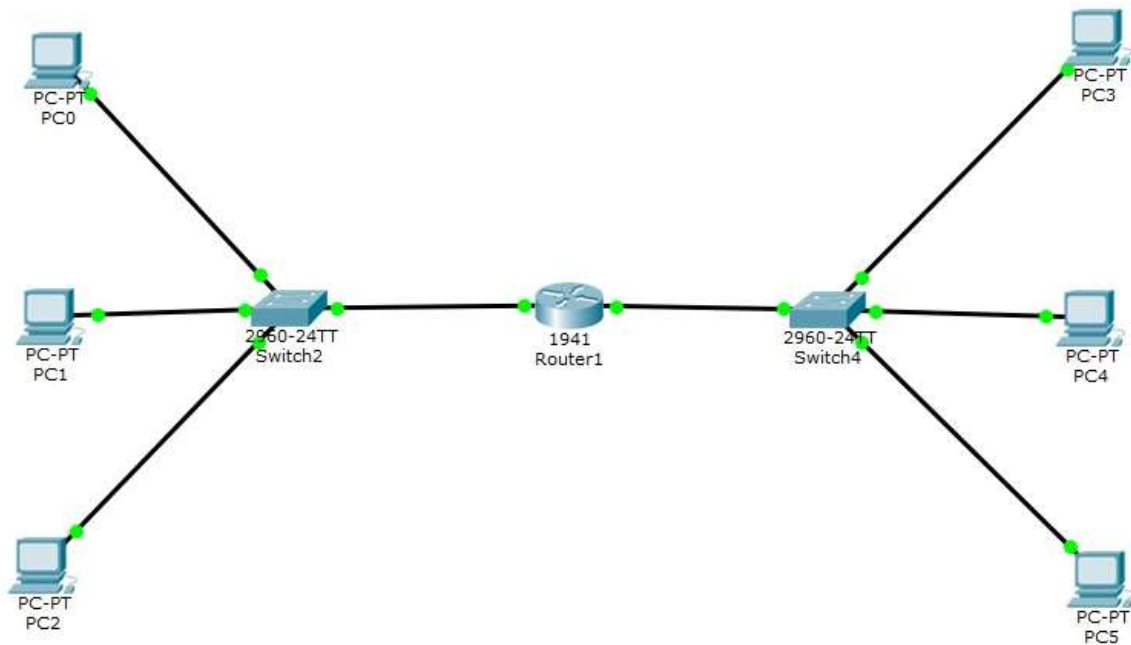
- PC 1: IP address 192.168.2.2, gateway 192.168.2.1
- PC 2: IP address 192.168.2.3, gateway 192.168.2.1
- PC 3: IP address 192.168.2.4, gateway 192.168.2.1

6. Test the network connection.

- Open a command prompt on one of the PCs on the left LAN.
- Ping one of the PCs on the right LAN.
- If the ping is successful, then the network connection is working properly.

7. Evaluate the network performance parameters.

- Open a command prompt on one of the PCs on the left LAN.
- Ping one of the PCs on the right LAN.
- Measure the latency, throughput, and packet loss.
- Repeat the steps above for different combinations of PCs on the two LANs.



EXP 5

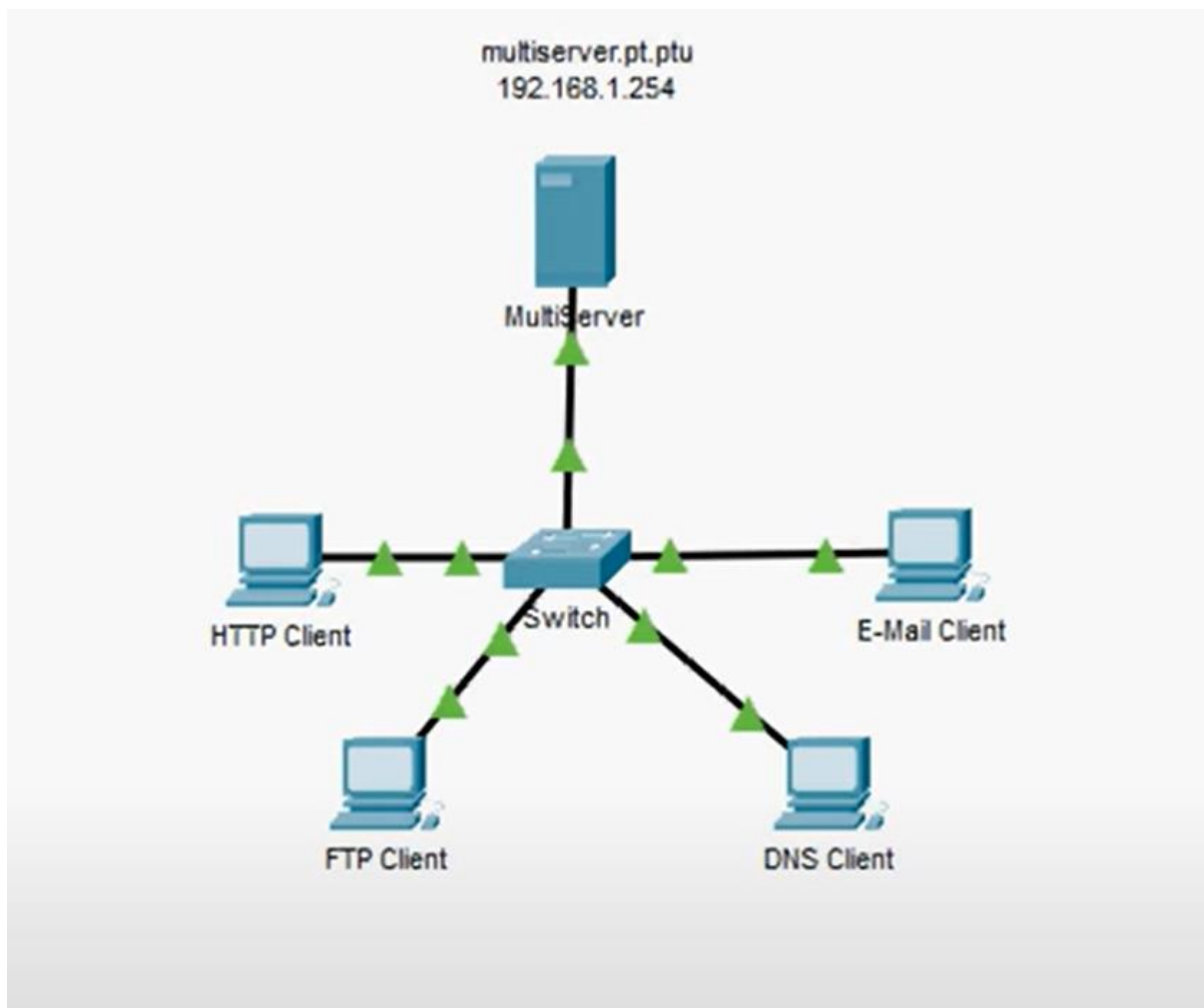
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Theory:

User Datagram Protocol (UDP) is a Transport Layer protocol. UDP is a part of the Internet Protocol suite, referred to as UDP/IP suite. Unlike TCP, it is an unreliable and connectionless protocol. So, there is no need to establish a connection prior to data transfer. The UDP helps to establish low-latency and loss-tolerating connections establish over the network. The UDP enables process to process communication.

For real-time services like computer gaming, voice or video communication, live conferences; we need UDP. Since high performance is needed, UDP permits packets to be dropped instead of processing delayed packets. There is no error checking in UDP, so it also saves bandwidth. User Datagram Protocol (UDP) is more efficient in terms of both latency and bandwidth.

Diagram:



Generating network for UDP transfer

Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following task to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the `ping -n 1 192.168.1.255` command. You are pinging the broadcast address for the LAN. The command option will send only one ping request rather than the usual four. This will take a few seconds as every device on the network responds to the ping request from MultiServer.
- Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

- Switch to Simulation mode.
- Click **HTTP Client** and open the **Web Browser** from the desktop.
- In the URL field, enter `192.168.1.254` and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- Minimize, but do not close, the **HTTP Client** configuration window.

Step 3: Generate FTP traffic.

- Click **FTP Client** and open the **Command Prompt** from the desktop.
- Enter the `ftp 192.168.1.254` command. PDUs will appear in the simulation window.
- Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

- Click **DNS Client** and open the **Command Prompt**.
- Enter the `nslookup multiserver.pt.ptu` command. A PDU will appear in the simulation window.
- Minimize, but do not close, the **DNS Client** configuration window.

Step 5: Generate Email traffic.

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Step 6: Verify that the traffic is generated and ready for simulation.

There should now be PDU entries in the simulation panel for each of the client computers.

Step 7: Examine multiplexing as the traffic crosses the network.

You will now use the **Capture/Forward** button in the Simulation Panel to observe the different protocols travelling on the network.